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1. Consider a Bernoulli population $Ber(p)$. Decide if the following hypothesis is simple or composite and why.

(a) $p = 0.7$

Simple

(b) $p > 0.5$.

Composite

2. Consider the Bernoulli population again. Suppose the null hypothesis $p = 0.7$ is tested against the alternative $p = 0.9$.

(a) Suppose that after the experiment, the outcome of the test is that the null hypothesis was rejected. What type of error, type I or type II, are we exposed to?

Type I

(b) Suppose x_1, x_2, x_3 is an observed sample. Intuitively, formulate a test in words. (ie, complete this sentence: I will reject the null hypothesis if _____.)

I will reject H_0 if $\bar{X}$, the sample average, is greater than some critical value I choose.

- (c) Continuing previous part. Suppose the test is $\bar{x} > 0.6$. If 1, 0, 1 is observed, what is the test outcome? Reject
- (d) Compute the probability of committing type I error if the test used is the one in the previous part.

$$\begin{array}{l} \text{Rejection} \\ \text{Space} \end{array} \left\{ \begin{array}{l} 111 \\ 011 \\ 101 \\ 110 \end{array} \right. \begin{array}{l} + (.7)^3 \\ + (.7)^2(.3) \\ + (.7)^2(.3) \\ + (.7)^2(.3) \end{array} = .49 + 1.6 + .784$$